

Autonomous Traffic-Lightless Intersection Control

1. Safety requirements

Collision Prevention

- Two conflicting vehicles shall never occupy the intersection simultaneously.

Mutual Exclusion

- Opposite unsafe directions shall never receive permission together.

Emergency Handling

- The system shall enter emergency mode if:
 - communication fails
 - sensor failure occurs
 - vehicle behavior becomes inconsistent

Deadlock Prevention

- The system shall never reach a state where all vehicles wait forever.

Safe Distance

- Vehicles shall maintain minimum separation distance.

Fail Safe Operation

- In case of controller failure:
 - All vehicles must stop before entering the intersection.

Priority Safety

- Emergency vehicles shall receive priority safely.

Safety_Requirements

<<requirement>> SR1
ID: SR1
Title: Collision Avoidance
Text: No two vehicles shall occupy the same conflict area or collision block simultaneously.

<<requirement>> SR2
ID: SR2
Title: Cross-Collision Point Safety
Text: Vehicles from conflicting directions shall not reach the same collision point at the same time.

<<requirement>> SR3
ID: SR3
Title: Permission Before Entry
Text: Vehicles shall only enter the intersection after receiving crossing permission from the intersection coordination unit.

<<requirement>> SR4
ID: SR4
Title: Minimum Vehicle Separation
Text: Vehicles traveling in the same lane shall maintain safe separation distance.

<<requirement>> SR5
ID: SR5
Title: Safe Turning Speed
Text: Vehicles performing left or right turns shall reduce speed to safe turning limits.

<<requirement>> SR6
ID: SR6
Title: Communication Failure Safety
Text: A disconnected vehicle without valid permission shall stop before the stop line.

<<requirement>> SR7
ID: SR7
Title: Sensor Failure Safe Mode
Text: If roadside or vehicle sensors become unreliable, the system shall switch to a safe operational mode.

<<requirement>> SR8
ID: SR8
Title: Emergency Vehicle Priority Safety
Text: Emergency vehicles shall receive priority crossing while maintaining safety for all other road users.

<<requirement>> SR9
ID: SR9

2. Timing requirements

Response Timing

- The controller shall respond to a vehicle request within a bounded time.
 - Example:
 - response time ≤ 2 seconds

Crossing Constraints

- A vehicle shall clear the intersection within a maximum crossing time.
 - Example:
 - crossing time ≤ 5 seconds

Communication Timing

- Vehicle status updates shall be transmitted periodically.
 - Example:
 - every 100 ms

Timeout Handling

- If a vehicle does not respond within a timeout interval, the controller shall revoke permission.
 - Example:
 - timeout = 3 seconds

Scheduling

- Only one conflicting traffic flow may be active at a time.
- Safe switching delay shall exist between permissions.

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